

Quantum Error Mitigation

Recent progress

Architecture · realistic noise · the PEC-bias artifact · readout & TREX

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Summary

- 1 - Backend / frontend separation: the qem package
- 2 - New noise models: Realistic Thermal Relaxation, Mixed Unitary & their composite
- 3 - PEC's positive-bias artifact: Aer's frozen RNG seed
- 4 - Readout errors & TREX mitigation
- 5 - Time varying noise



Architecture: backend / frontend split

From one long notebook to a reusable package

Frontend — notebooks

DZNE_VS_PEC_mitiq.ipynb

Composite_noise_DZNE.ipynb

Composite_readout_TREX.ipynb

One setup cell: `from qem import *`

Each experiment = 1 line:

```
cached_benchmark(SPEC_...)
```

Backend — qem/ package

`config.py` global parameters

`circuits.py` mirror benchmark

`kraus.py` channel algebra

`noise_models.py` noise + PEC reps

`pec_core.py` QPR / sampling

`benchmark.py` `run_benchmark`

`readout.py` readout + TREX

`cache.py` `cached_benchmark`

Why split : the advantages

- **No cell-order dependency** : import once, use anywhere
- **Readable notebooks** : machinery hidden; each cell reduced to a 1-line `run_benchmark(...)` call
- **Reuse** : the same backend powers 3 notebooks + a dozen smoke tests (`smoke_*.py`).
- **Disk cache** (`qem/cache.py`) : results pickled, keyed by params + source of the noise builder; editing a model auto-invalidates.
- **Maintainable** : clear separation: global params in `config.py`, per-model params beside their spec in `noise_models.py`.

New noise : Thermal Relaxation (realistic)

Per-gate noise from real device timings

- **thermal_relaxation_error**($T_1 = 100 \mu\text{s}$, $T_2 = 80 \mu\text{s}$, **gate_time**) : one channel per gate, scaled by its own duration.
- **Durations** (IBM-style): rz / id = 0 (virtual $\rightarrow$ no noise) · sx / x / h / sxdg $\approx 35 \text{ ns}$ · cx $\approx 400 \text{ ns}$ (dominant relaxation).
- **Zero-temperature**, $T_2 \leq T_1 \rightarrow$ exactly : $\mathcal{E}_{\text{thermal}} = \text{AD}(\gamma_{ad}) \circ \text{PD}(\gamma_{pd})$

$$\gamma_{ad} = 1 - e^{-t/T_1} \qquad \gamma_{pd} = 1 - e^{-2t/T_2 + t/T_1}$$

Verified to machine precision ($\|S_{\text{AD} \circ \text{PD}} - S_{\text{qiskit}}\|_{\infty} \approx 1\text{e-15}$) $\rightarrow$ PEC inverts the exact channel $\rightarrow$ unbiased.

New noise : Mixed Unitary (biased Pauli)

Coherent / dephasing residual after calibration

- A Pauli channel IS a mixed-unitary channel (a convex mixture of Pauli unitaries).

$$\mathcal{D}(\varepsilon) = (1 - \varepsilon) [I] + \varepsilon \left(\frac{\eta}{\eta + 1} [Z] + \frac{1}{3(\eta + 1)} ([X] + [Y] + [Z]) \right)$$

- $\eta = 0 \rightarrow$ pure depolarizing · $\eta \rightarrow \infty \rightarrow$ pure dephasing · $\eta > 0$ adds the X/Y/Z asymmetry plain depol lacks.
- **Realistic params:** $\varepsilon_1 q = 1e-3$, $\varepsilon_2 q = 5e-3$, $\eta = 4$ (biased toward Z / dephasing, as on $T_2 < T_1$ devices).
- Mitiq ships an analytic QPR for this channel $\rightarrow$ PEC needs no LP machinery. Unital $\rightarrow$ fixed asymptote for ZNE.
- Both models act on physical gates only (rz / id are virtual $\rightarrow$ no error).

Composite hardware model

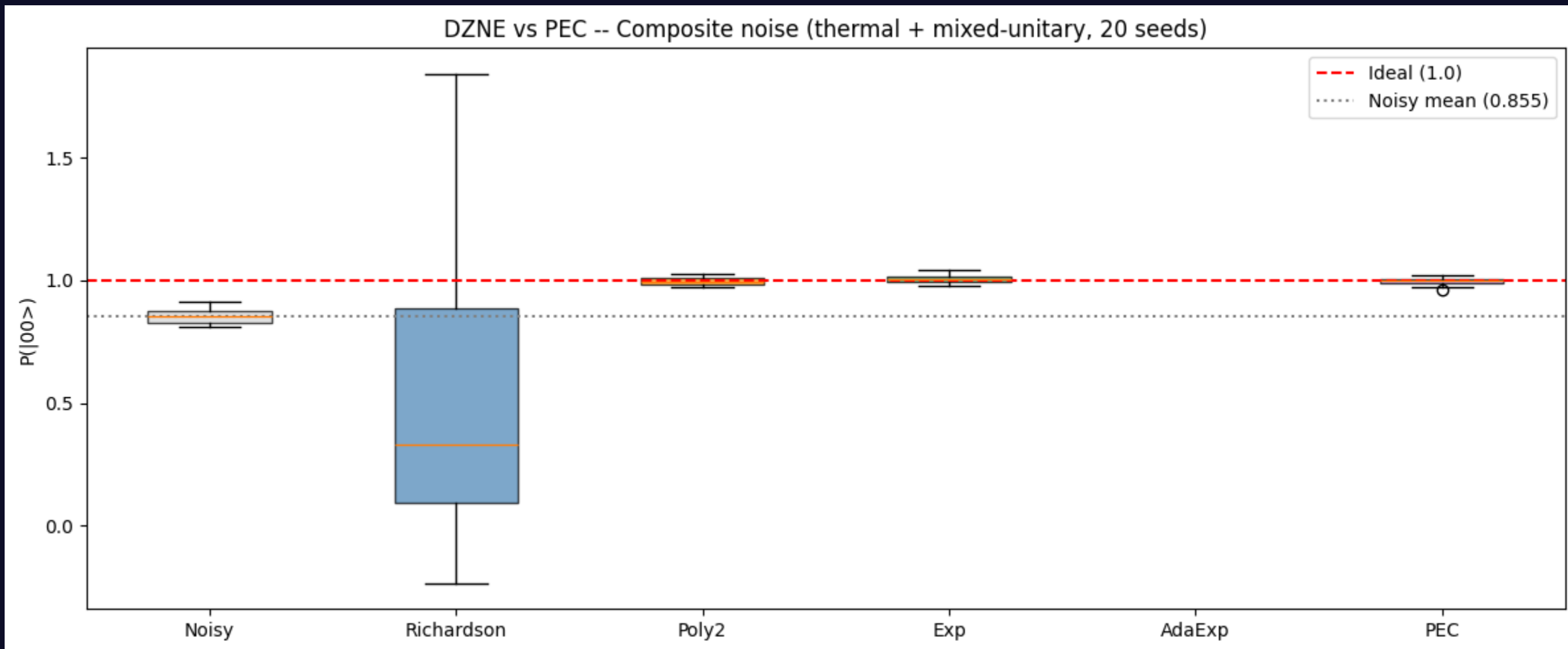
Thermal Relaxation ◦ Mixed Unitary : closest static approximation to hardware

- Two physically-distinct mechanisms, stacked per gate:
 - incoherent T_1 / T_2 relaxation : **ThermalRealistic (non-unital)**
 - biased-Pauli residual left after calibration : **Mixed Unitary**

$$\text{gate channel} = \text{ThermalRealistic}(\gamma) \circ \text{MixedUnitary}(\varepsilon, \eta)$$

- No standalone depolarizing: Mixed Unitary already contains it — its $1/(3(\eta+1))([X]+[Y]+[Z])$ term IS depolarizing ($\eta = 0$). A separate depol would double-count the isotropic Pauli error.
- Physical gates only (sx / sxdg / x / h / cx); rz / id virtual $\rightarrow$ no noise. CX: $(\text{th} \otimes \text{th}) \circ (\text{mu} \otimes \text{mu})$.

Composite: results



PEC's positive bias : a hidden artifact

Aer's frozen RNG seed

- Symptom: the PEC estimator was biased $\approx +0.06$ to $+0.13$, with strongly inflated variance.
- Root cause: `run_benchmark` shared ONE `AerSimulator(seed_simulator = 42)`. PEC calls `backend.run()` thousands of times; a frozen seed restarts every run from the SAME RNG stream $\rightarrow$ noise realizations correlated across PEC importance samples $\rightarrow$ biased mean + inflated variance.
- Diagnosis by elimination (smoke tests):
 - reps exact to $1e-7$ · shot vs density-matrix identical · PEC with a RANDOM seed unbiased at every shot budget.

The fix: one seedless backend for all methods

- A single AerSimulator with NO seed, shared by every method (noisy / ZNE / PEC) → independent noise per run, honest variance, all methods on equal footing. Reproducibility comes from the disk cache instead.

Results : 5 seeds, composite noise

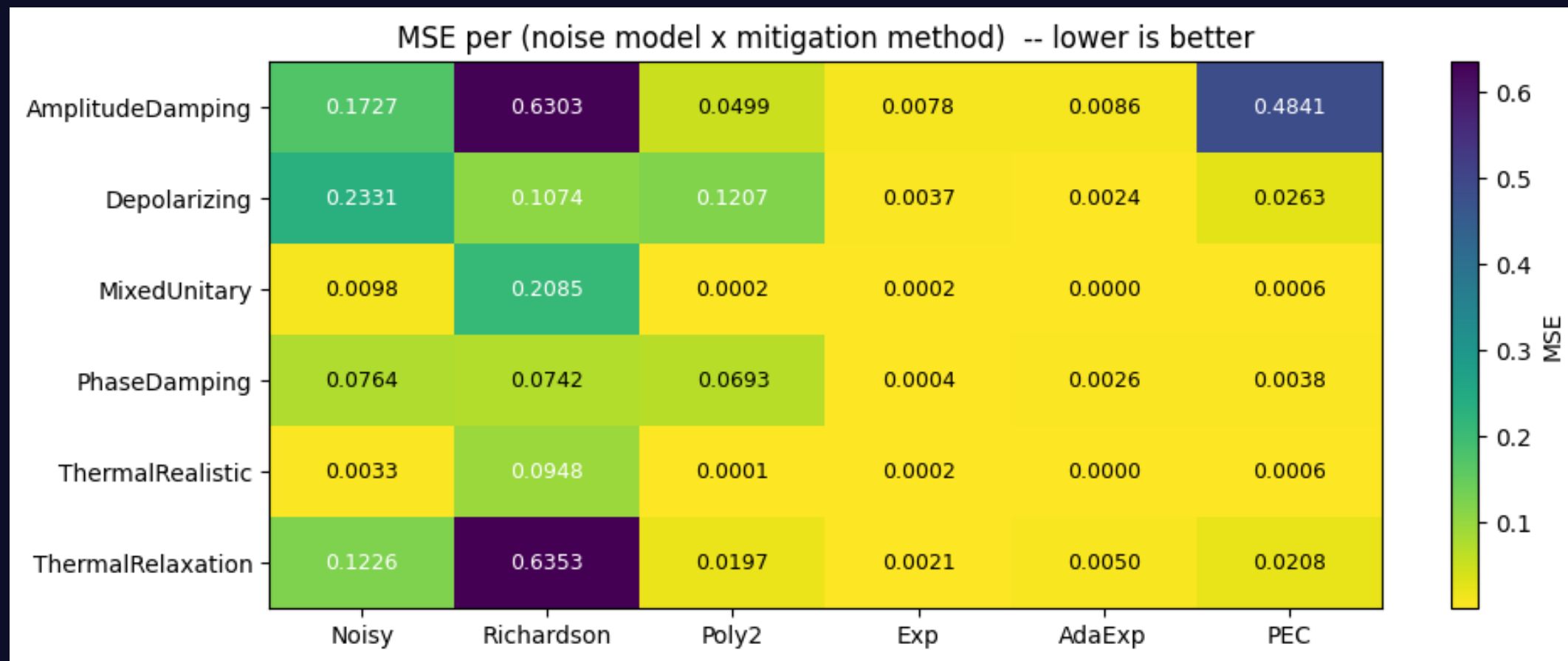
PEC bias: +0.129 → +0.004

Variance: 0.00415 → 0.00015

MSE → ~5e-5 → PEC now competitive with AdaExp

The artifact likely inflated PEC across ALL noise models : their earlier PEC numbers were overstated.

DZNE vs PEC results



PEC overhead is higher in the toy noise models

Readout errors

A classical bit-flip QEM cannot reach

- Readout error = a classical bit-flip on the measured bit, independent of gate noise:
 - $p_0 = P(\text{measure } 1 \mid \text{true } 0)$ $p_1 = P(\text{measure } 0 \mid \text{true } 1)$
- NOT a gate channel $\rightarrow$ no PEC representation; neither ZNE nor PEC correct it.
- Realistic (IBM Eagle / Heron): $p_0 = (0.010, 0.013)$, $p_1 = (0.025, 0.030)$.
 - Asymmetric — T_1 relaxation during the $\sim 1\text{--}2 \mu\text{s}$ measurement makes $p_1 > p_0$ ($\sim 1.8\text{--}2.1$ % assignment error).

TREX: Twirled Readout Error eXtinction

- Readout X-twirling symmetrizes each qubit's $(p_0, p_1) \rightarrow \bar{p} = (p_0 + p_1) / 2$.
- Then each Z carries a clean eigenvalue:

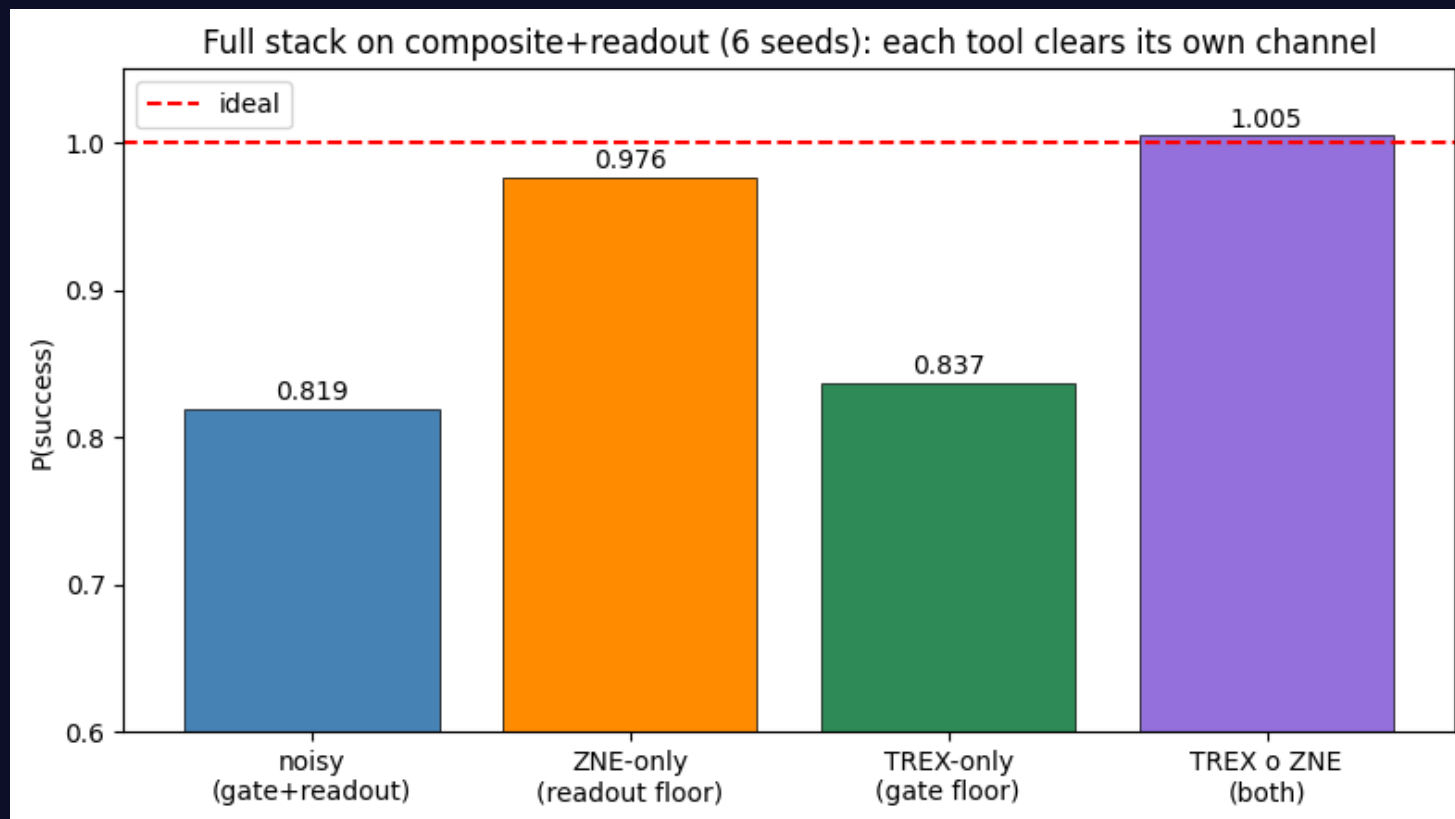
$$\lambda_i = 1 - 2\bar{p} = 1 - (p_{0,i} + p_{1,i})$$

$$\lambda_w = \prod_{i \in w} \lambda_i$$

- TREX calibrates λ_w on $|0\dots 0\rangle$ and divides $\langle Z_w \rangle$ by it.
 - Twirling the calibration too is what removes the asymmetry bias (raw calib sees only p_0).
- Implementation: `mitiq.experimental.trex`.

Orthogonal complementarity (full stack)

Composite gate noise + readout — $P(\text{target})$, ideal = 1.0



Readout mitigation and gate-noise QEM are orthogonal : both are needed.

Time varying noise model

Real devices drift :

- T1, T2, readout errors and more generally, noise channels change over time
- There are several ways to model this phenomenon, depending on the time scale of the variation :

Variation during
circuit execution

Variation
between circuit
executions

What I implemented

Consequences :

- PEC relies on an exact noise **representation** : a bias will appear
- ZNE assumes monotone amplification : it should be robust

Modeling drift : Ornstein-Uhlenbeck (OU)

- Temporal modulation of the existing static model parameters
- One OU process per parameter : $\{T_1\}$ $\{T_2\}$ $\{\varepsilon\}$ {readout}

$$dX_t = \theta (\mu - X_t) dt + \sigma dW_t$$

Noise
parameter

Reversion
speed

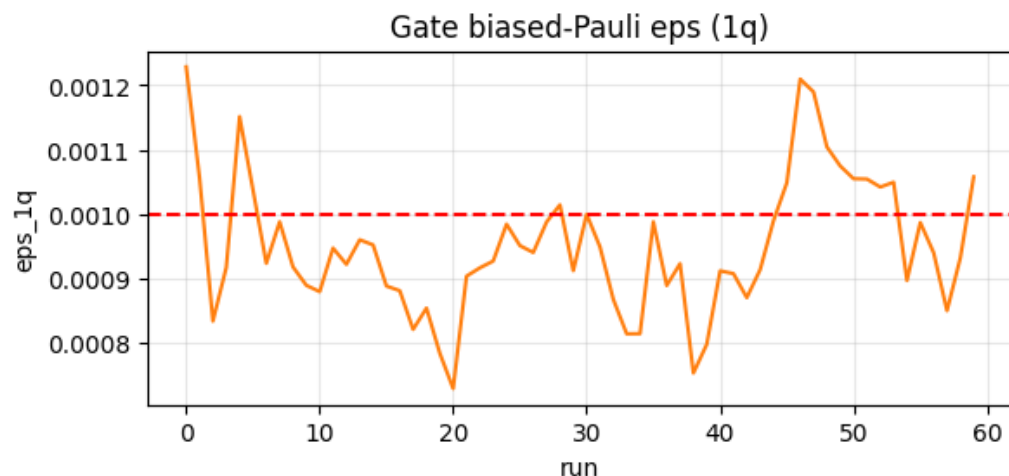
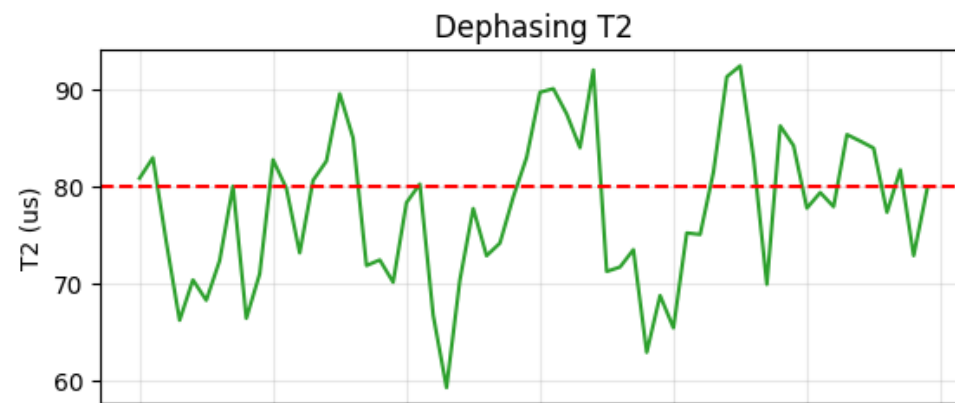
Equilibrium
level

Diffusion
coefficient

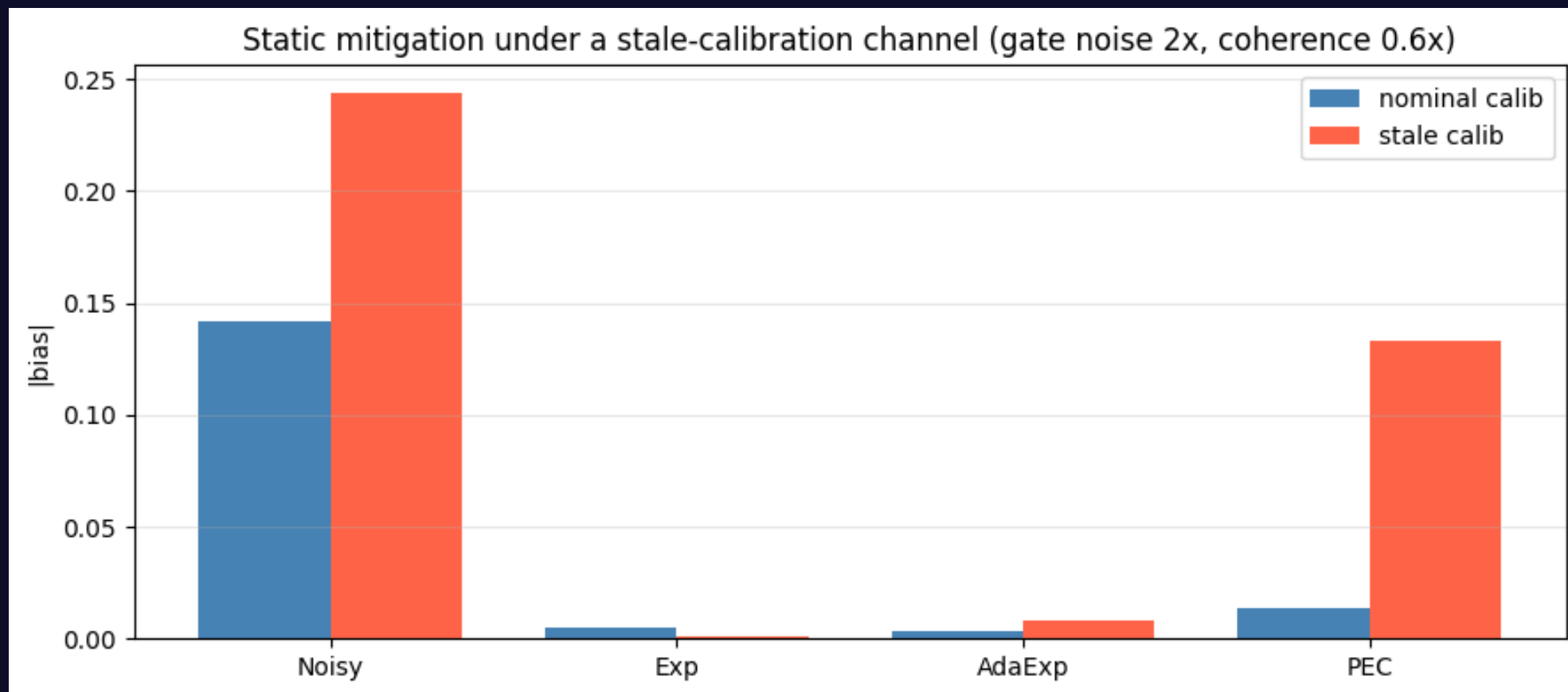
Standard
Brownian
movement

Modeling drift : Ornstein-Uhlenbeck (OU)

Inter-run drift: 4 independent OU clocks (dashed = nominal), $\sigma=0.15$, $\tau=5$
this notebook uses T1/T2/eps (gate); readout drives the TREX sibling



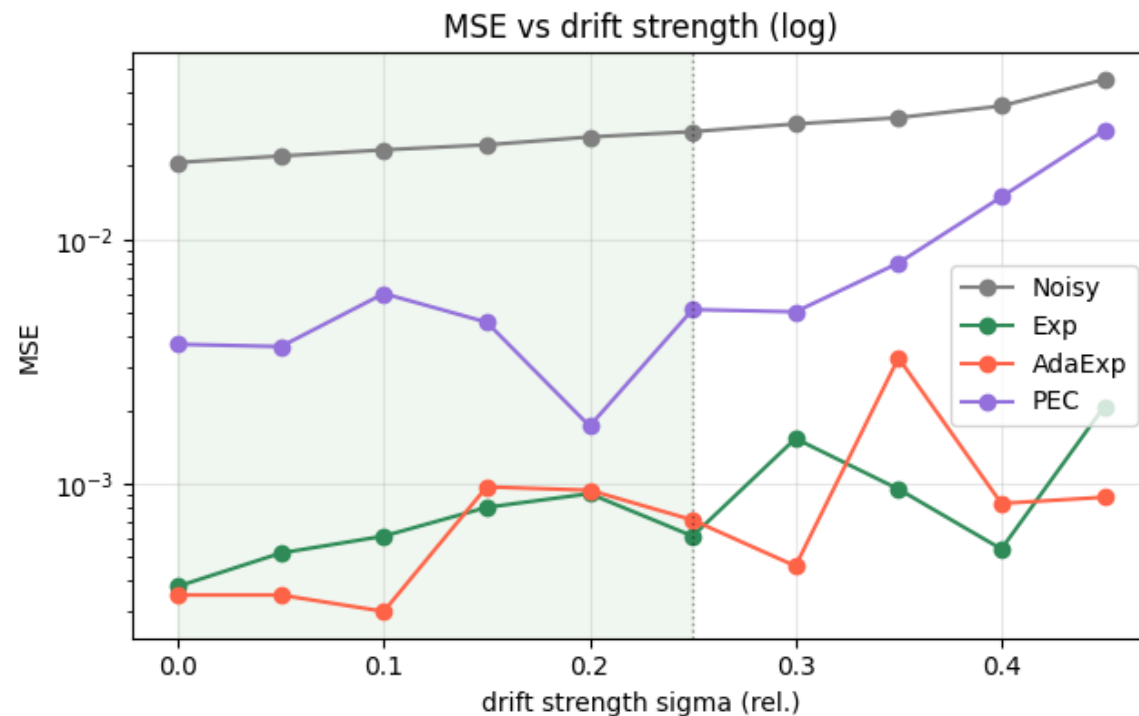
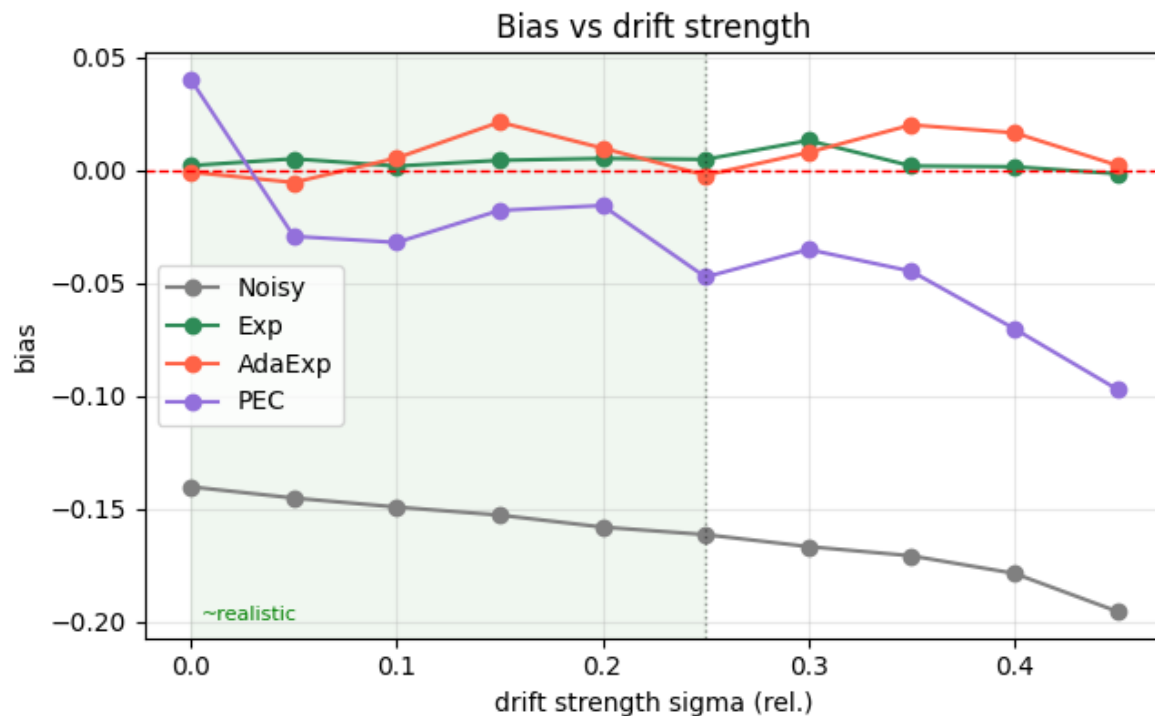
Stale calibration



- PEC bias explodes (9 times stronger) when the noise is different from the initial calibration

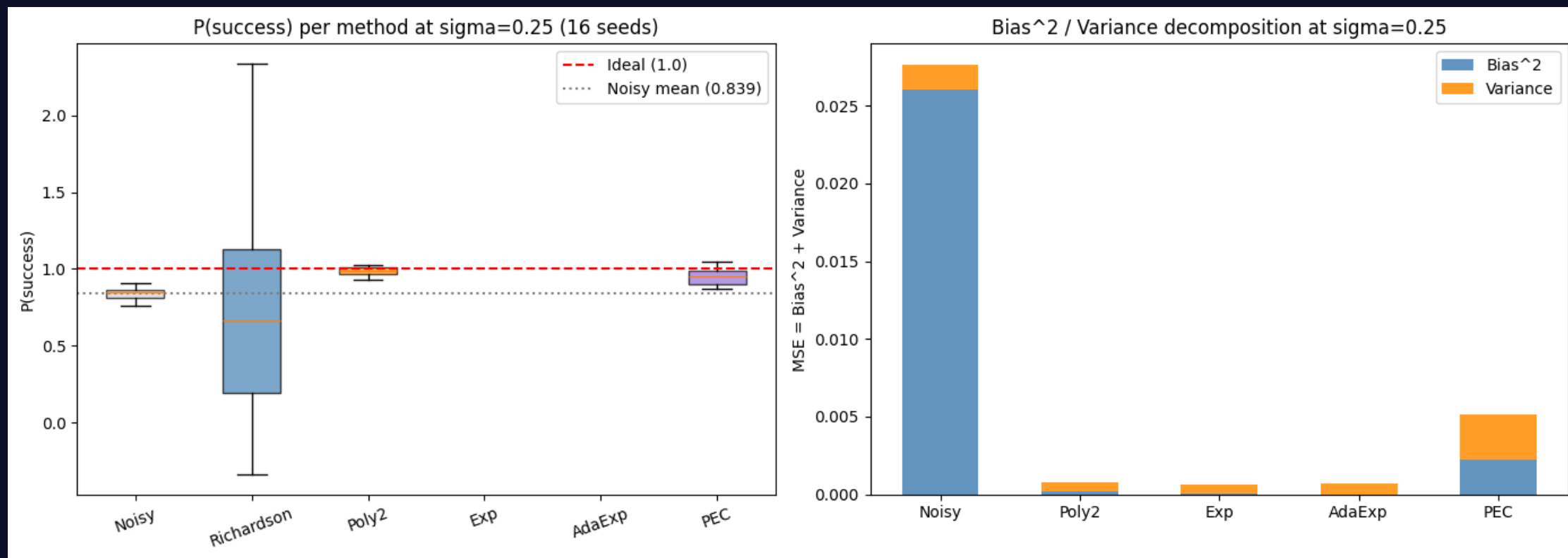
Gate channel : ZNE vs PEC

Static mitigation degrades with inter-run drift (16 seeds, budget 12,000)



- Gate only channel (thermal realistic + mixed unitary) no readout errors yet
- PEC becomes more biased as the noise drifts further away from the exact representation, expected

Bias vs Variance



- Under symmetric drift, the bias averages out, appearing as **variance**
- Using static methods on time varying noise is a bottleneck : the next goal is to **relearn the noise**

Takeaways

- A **reusable** qem backend turned the study into **1-line, cached experiments** across several notebooks.
- **Two hardware-flavoured noise models added** : Thermal Relaxation (exact AD•PD) and biased-Pauli Mixed Unitary, both with **unbiased PEC**.
- The PEC "positive bias" was an Aer frozen-seed artifact, not a method flaw; seedless runs make **PEC competitive with AdaExp**.
- Readout errors sit outside QEM's reach : TREX closes the gap; composed with ZNE it recovers the ideal.
- **Time dependant parameters** added to the composite (gate) noise model to quantify how the static methods degrade
- **Next** : include the time dependent parameter in readout errors, mitigate them with TREX and begin to shift towards **noise model learning** and **PEA**.

Any questions or feedback?

Thank you for your attention and insights

